

Session Objectives

- Demonstrate the components of the history taking and physical examination of the Spine
- Describe subjective and objective findings relating to the examination of the spine
- Distinguish Normal vs. Abnormal examination findings
- Recall the appropriate way to document common normal and abnormal findings of the history and physical Examination
- Incorporation of OMM in a Case Study

The Health History

- Common or Concerning Symptoms
 - Joint pain: articular or extra-articular, acute or chronic, inflammatory or noninflammatory, localized or diffuse, monoarticular or polyarticular
 - Joint pain: associated constitutional symptoms and systemic manifestations from other organ systems
 - Fever, chills, rash, weight loss, weakness
 - Neck pain
 - Low back pain

History- Low Back Pain

- Is the pain on the midline or off the midline
- Is there radiation into the buttock or lower extremity?
- Associated numbness or paresthesia?
- Leg pain that resolves with rest and/or lumbar forward flexion
- Is there an associated bowel or bladder dysfunction?
- “Red Flags” for serious underlying systemic disease

History Taking- Back Pain

- Assessed decreased or limited movement
- Timing
 - Acute vs. Chronic
 - What aggravates or relieves the pain
- Systemic Features
 - Fever? Chills? Rash? Fatigue? Anorexia? Weight loss? Weakness?

“Red Flags” for Low Back Pain

- History of IVDA, addiction, or immunosuppression
- Presence of active infection or HIV
- Long-term steroid therapy
- Saddle anesthesia, bladder or bowel incontinence
- Neurologic symptoms or progressive neurologic deficit

Common Review of System

- Peripheral Vascular: No varicose veins
- MSK: Any other joint pain, muscle pain/weakness
- Neuro: Numbness / Tingling / Weakness / Saddle Anesthesia / loss of bowel and bladder control
- Endocrine: Excessive thirst / hunger / urination
- GU: F / U / D; Dysuria; Discharge
- Psychiatric: Depression or anxiety

Cervical Spine- PALPATION

- Sitting or standing position
- Spinous processes
 - Note pain, swelling, step-offs
 - Any cervical spine with spinous processes that show either a step-off or a significant difference in interspinous distance needs radiographs
- Laterally and palpate the musculature in the posterior neck
- Suboccipital fossae noting tenderness and somatic dysfunction
- Greater occipital nerves may be tender in patients with chronic headaches

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine- Range of Motion

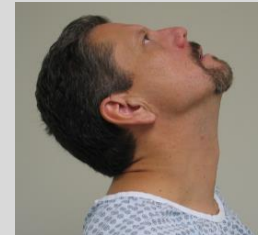
- Neck is the most mobile of the spine
 - Flexion- “Bring your chin to your chest”
 - Extension- “Look up at the ceiling”
 - Rotation- “Look over one shoulder, and then the other”
 - Lateral bending- “Bring your ear to your shoulder”
- Cervical Flexion and extension → OA joint
- Cervical Rotation → AA joint
- Lateral bending → C2-C7

Sources:

1. Bates’ Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Cervical Spine- Range of Motion Testing

- Active Range of Motion
 - Instructions to Patient
 - Extension → “Look up at the ceiling”
 - Flexion → “Bring your chin to your chest”



Cervical Spine- Range of Motion Testing

- Active Range of Motion
 - Instructions to Patient
 - Rotation → “Look over one shoulder, then the other”
 - Lateral bending → “Bring your ear to your shoulder, then to the other”
(Make sure they don’t raise the shoulder)



Cervical Spine Examination

- Lhermitte's Sign
 - A.K.A. Barber's chair phenomenon
 - While sitting on the exam table and attempting to actively hold their head upright, the patient complains of electrical shock sensations down the spine, arms, or legs
 - Indicates possible Spinal stenosis (Rule out Multiple Sclerosis, ALS, Arnold-Chiari Malformation)

Sources:

1. Bates' Guide to the Physical Examination and History Taking; 12th Ed. – Chapter 16
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

Axial Compression Test

- Examination
 - Standing behind
 - Perform test once
 - Examiner places both hands on top of patient's head
 - Exerts a downward force with both hands
- Positive test
 - Pain and/or neurological symptoms of the cervical spine and/or upper extremity are reproduced with the compressive force
- Indicates
 - Cervical Nerve Impingement

Foraminal Distraction Test

- Examination
 - Standing at side of table facing the patient
 - Perform test once
 - Examiner places one hand under the patient's chin and the other under the occiput, then applies an upward traction force on the cervical spine
- Positive test
 - Pain and/or neurological symptoms of the cervical spine and/or upper extremity are reduced with the traction force
- Indicates
 - Cervical Nerve Impingement

Spurling's Test

- Examination
 - Test for foraminal compression on a nerve root
 - Examiner places one hand on the top of the patient's head and the other on patient's shoulder at the affected side
 - Examiner proceeds to side bend and rotate the patient's neck TOWARD the side of their pain
 - Adds slight extension of the patient's neck
 - The examiner exerts a downward force on the patient's spine
- Positive test
 - Pain radiating down the arm to which the head is rotated
- Indicates
 - Cervical radiculopathy

Sources:

1. ICC Guide
2. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 9

Cervical Spine- Common Conditions

- Soft Tissue Injuries / Conditions
 - Cervical strain, cervical sprain
 - Most Common
 - Soft tissue activity threshold is surpassed
 - Usually self-limited
 - Cervical Somatic dysfunction
 - Headache aka “cephalgia”
 - Stingers / Burners

Cervical Spine- Common Conditions

- Hard Tissue Injuries / Conditions
 - Osteoarthritis
 - Common condition whereby the multiple joints and vertebral bony components develop the classic changes associated with OA
 - Degenerative disc disease
 - Wear and tear phenomena that occur across the disc-vertebrae interface
 - Associated with aging and the inherent loss of water content within the disc itself

Cervical Spine- More Urgent

- Fracture
 - Moderate forces are at play
 - Pain and dysfunction are disproportional to the physical findings
 - Pinpoint tenderness
 - Neurologic finding
- Herniated Disc
- Infection
- Inflammatory Condition
- Malignancy
- Myelopathy
- Thoracic outlet Syndrome
- Vascular Emergency

Source: Stockard, Alan, *Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine*; Chapter 1
Childress MA, Stueck SJ. Neck Pain: Initial Evaluation and Management. *Am Fam Physician*. 2020 Aug 1;102(3):150-156.
PMID: 32735440.

Thoracic-Lumbar Spine- Range of Motion

- **Flexion (Active)**
 - Examiner instructs patient to bend forward and touch toes, noting smoothness and symmetry of movement, the range of motion, and the curve in the lumbar area
- Extension
- Rotation
- Side Bending

Thoracic-Lumbar Spine- Range of Motion

- Flexion (Active)
- **Extension (Active)**
 - Examiner places one hand on the posterior superior-iliac spine, fingers pointed to mid-line to support the patient's back, then instructs patient to bend back as far as possible
- Rotation
- Side Bending

Thoracic-Lumbar Spine- Range of Motion

- Flexion (Active)
- Extension (Active)
- **Rotation (Passive)**
 - Examiner places one hand on patient's iliac crest (to stabilize the pelvis) and the other hand on patient's opposite shoulder.
 - The examiner rotates patient's trunk by pulling the shoulder and then the hip posteriorly
 - Examiner changes his/her hand positions and repeats procedure for the opposite side
- Side Bending

Thoracic-Lumbar Spine- Range of Motion

- Flexion (Active)
- Extension (Active)
- Rotation
- **Side Bending** (Active)
 - Examiner stabilizes patient's pelvis by placing one hand on the patient's iliac crest and then instructs patient to lean as far as possible on the ipsilateral side.
 - Examiner changes his/her hand position and repeats procedure for the opposite side

Scapular Elevation

- Examination
 - The examiner stands behind the patient and places his or her hands on each acromion.
 - With arms resting at the side, the patient shrugs his or her shoulders against examiner resistance
- Positive Test
 - Weakness or pain against resistance
- Indicates
 - Tendonitis (of trapezius and/or levator scapulae muscles), cervical injury, or cranial nerve damage (Spinal Accessory nerve, CN XI)

Scapular Protraction (Winging)

- Examination
 - The patient pushes against a wall with both hands with the feet farther away from the wall than the shoulder.
- Positive Test
 - Scapular winging, pain, and/or weakness during the maneuvers
- Indicates
 - **Serratus anterior muscle** weakness. Lateral winging can be generated by trapezius and rhomboid paralysis

Adson's Test

- Examination
 - The examiner locates the radial pulse.
 - The patient rotates their head toward the tested arm and lets the head tilt backwards (extends the neck) while the examiner abducts/extends the arm.
 - The patient is asked to inhale and turn the head toward the affected side
- Positive Test
 - The disappearance or diminution of the pulse or provocation of symptoms
- Indicates
 - Thoracic outlet syndrome, usually related to the scalene musculature

Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 5
2. PowerPoint Presented to Class of 2021- *Foundations of Osteopathic Medicine*- OMM Lab #3 – “Osteopathic Evaluation of the Shoulder” presented August 23rd and 24th, 2017

Adam's Forward Bend Test

- Examination
 - Patient stands with feet together
 - Examiner instructs patient to bend spine forward while keeping their feet together, knees straight and their hands hanging at their side
- Positive Test
 - Presence of unilateral rib cage elevation or unilateral lumbar paravertebral muscle prominence
- Indicates
 - The curve of structural scoliosis is more apparent when the patient forward bends

Normal

Abnormal

Stork Test

- Examination
 - Single leg weight-bearing hyperextension test where the patient stands on one leg and then leans backward extending the lower back
- Positive Test
 - The patient experiences pain in the low back, as this stresses the posterior elements of the spine on the ipsilateral side
- Indicates
 - A possible pars defect / stress fracture, spondylolysis, or facet pathology

Straight Leg Raise Test

- Examination
 - Patient lying supine and relaxes his/her leg
 - The leg with knee extended is passively raised off the examination table
 - Bilateral Exam. Uninvolved side first.
- Positive Test
 - A radicular pain emanating from the low back following the sciatic nerve distribution, and usually radiating below the knee on the ipsilateral side
- Indicates
 - Nerve root impingement

Sources:

1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 6
2. PowerPoint Presented to Class of 2021- *Foundations of Osteopathic Medicine*- OMM Lab #11 – “Osteopathic Evaluation of the Hip” presented November 1st and 2nd, 2017

Well (Opposite) Straight Leg Raising

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- Examination
 - Patient raises his/her straight leg on the uninvolved side
- Positive Test
 - Pain is experienced on the side opposite that being raised
- Indicates
 - The test is positive for possible space occupying lesion if pain occurs on the opposite side

Source: ICC Guide

Hoover Test

- Examination
 - With the patient lying supine, the examiner places a hand under each heel.
 - The patient is then asked to lift the affected leg off the table.
 - A normal response is for the patient to push downward with the contralateral heel
- Positive Test
 - No downward force is felt, the patient is not giving enough effort when the patient is raising the involved leg
- Indicates
 - The patient may be malingering, or seeking a secondary gain

Lumbar Spine- Differential Diagnosis

- Lumbar Strain/Sprain
- Herniated nucleus pulposus
- Spinal Stenosis
- Spondylolisthesis
- Spondylolysis
- Spondylosis (degenerative disc disease)
- Compression fracture

Case Study

- Allergies: PCN (rash)
- Medications: Selegiline 5mg bid, amlodipine 5mg daily, claritin 10mg as needed
- PMHx: Parkinson's Disease (diagnosed 10 years ago), Hypertension (since age 55)
- PSHx: Appendectomy (age 30)
- FH: Mother, deceased at age 92, HTN; Father, deceased at age 87, CAD
- Social history: Denies history of tobacco or recreational drug use. Denies alcohol use. Retired Accountant. Limited exercise secondary to pain. Diet includes fruits and vegetables.
- HCM: Last colonoscopy 10 years ago was normal. He received the flu vaccine and pneumovax this year. Last physical exam was 3 months ago.

Case Study

- Physical Exam:
 - Vitals: BP- 124/80, HR 74, RR 14, Temp. 97.9, Weight- 165 pounds, Height- 72 inches, BMI- 22.38, Pulse Ox- 98% room air
 - General: Awake, alert, and oriented x 3, No acute distress, pleasant affect
 - Head: Normocephalic, atraumatic
 - Extremity: No clubbing / cyanosis / edema, pulses 2+ bilaterally in both upper and lower extremities

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - MSK: Thoracic spine with + head forward and rounder shoulder posture. There is no vertebral spine tenderness. There is paraspinal muscle spasm presents bilaterally. There is increased Thoracolumbar spinal flexion. Range of motion extremely limited In all direction

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - OSE:
 - Sub-occipital hypertonicity, OA E SI Rr, SEVERITY: Moderate, Treated with: Sub-occipital release, BLT indirect, TX RESPONSE: Improved
 - Cervical region with hypertonicity of bilateral upper trapezius, C4, FRS left, Paravertebral muscle hypertonicity bilaterally, SEVERITY: Severe, Treated with: Facilitated Positional Release, Myofascial release, BLT indirect, TX RESPONSE: Improved

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Physical Exam:
 - OSE:
 - Upper extremity: Fascia restrictions in bilateral shoulders resulting in decreased range of motion, SEVERITY: Moderate, Treated with: Modified Spencer's Technique in supine position with Direct Fascial Release

Case Study

- Assessment:
 - Back pain with multifactorial etiology
 - Degenerative Disc Disease
 - Herniated Disc
 - Spondylosis
 - Multiple Somatic Dysfunctions (Cranial, Cervical, Thoracic, Lumbar)
 - Parkinson's Disease
 - Camptocormia

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Case Study

- Plan:
 - OMM treatment done today in office
 - Proper hydration and healthy nutrients
 - May apply heat as needed
 - Back stretches and ROM exercises advised
 - Continue current Physical Therapy for improvements in balance and gait
 - Tylenol / NSAIDs as needed for pain
 - Will consider getting repeat imaging if symptoms persists or worsens
 - Follow up in 1 week, sooner if symptoms worsens

Source: Real Case at NYIT Academic Health Care Center

Note: Age and non-pertinent information have been modified

Summary Slide

- Demonstrate the components of the history taking and physical examination of the Spine
- Describe subjective and objective findings relating to the examination of the spine
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